SWAR popcnt for 32-bit reg

```
mov     edx,eax
and     eax,0x55555555
shr     edx,0x1
and     edx,0x55555555
add     eax,edx

mov     edx,eax
and     eax,0x33333333
shr     edx,0x2
and     edx,0x33333333
add     eax,edx

mov     edx,eax
and     eax,0x0f0f0f0f
shr     edx,0x4
and     edx,0x0f0f0f0f
add     eax,edx

mov     edx,eax
and     eax,0x00ff00ff
shr     edx,0x8
and     edx,0x00ff00ff
add     eax,edx

mov     edx,eax
and     eax,0x0000ffff
shr     edx,0x10
and     edx,0x0000ffff
add     eax,edx
```

```
mov   edx, eax
and   eax, 0x55555555
shr   edx, 0x1
and   edx, 0x55555555
add   eax, edx
mov   edx, eax
and   eax, 0x33333333
shr   edx, 0x2
and   edx, 0x33333333
add   eax, edx
mov   edx, eax
and   eax, 0x0f0f0f0f
shr   edx, 0x4
and   edx, 0x0f0f0f0f
add   eax, edx
mov   edx, eax
and   eax, 0x00ff00ff
shr   edx, 0x8
and   edx, 0x00ff00ff
add   eax, edx
mov   edx, eax
and   eax, 0x0000ffff
shr   edx, 0x10
and   edx, 0x0000ffff
add   eax, edx
```

leaves space for add (is no overflow)

1 → 2 → 4 → 8 → 16

gives # of 1s in eax

32 = # of 1s (eax = 1111 ... 1111)

32 + 8 + 1 = 41?

ie popcnt / hamming weight for a 32-bit int

(& sweet)